# ShadowScan – Phase One: Explainable AI for Network Intrusion Detection

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## 1. Abstract

The growing complexity of network environments and the increasing number of connected devices have made cybersecurity monitoring more challenging. Security alerts originate from different sources and are often scattered across systems, making it difficult for security personnel to examine them together, understand their significance and make informed decisions. This is particularly challenging for personnel without specialised cybersecurity expertise. Although machine learning-based Intrusion Detection Systems (IDS) can identify potentially suspicious network activity, their predictions may provide limited insight into the factors behind individual classifications. This creates a need for more interpretable approaches to cybersecurity monitoring and threat assessment.

This study forms Phase One of ShadowScan, a broader research and development initiative focused on explainable, AI-assisted cybersecurity monitoring and decision support. It investigates the use of Explainable Artificial Intelligence (XAI) to make network intrusion detection predictions more understandable and explore their potential role in supporting human assessment.

Experiments were conducted using the RT-IoT2022 dataset, comprising 123,117 observations across 12 traffic categories. Following preprocessing, 93 features were retained. Decision Tree and Random Forest classifiers were compared, with Random Forest selected for further analysis. It achieved 99.83% accuracy, a macro F1-score of 96.22% and balanced accuracy of 96.57% on the held-out test dataset. SHapley Additive exPlanations (SHAP) was integrated to examine global, class-wise and individual predictions, feature contributions, explanation consistency and model misclassifications. A Streamlit-based prototype was also developed to present predictions and their explanations through a single interface.

To explore how people interpret these explanations, a screenshot-based comprehension survey was conducted. At the time of reporting, 11 responses had been received. Data collection and analysis remain in progress.

The study examines the interpretability of machine learning-based threat detection, the consistency of SHAP explanations and their potential to support human assessment within a centralised cybersecurity monitoring framework. The findings provide an initial technical and analytical foundation for the continued development of ShadowScan.

## 2. Keywords

**Network Intrusion Detection; Explainable Artificial Intelligence; Machine Learning; SHAP; Random Forest; Cybersecurity Monitoring; Human Decision Support.**

## 3. Introduction

### 3.1 Background

The growing number of connected devices and the increasing complexity of network environments have expanded the volume and variety of cybersecurity events that organisations must monitor. Detecting suspicious activity is only one part of the challenge. Security alerts originate from different sources and are often scattered across systems, making it difficult for security personnel to examine them together and understand their significance. This becomes even more challenging for those without specialised cybersecurity knowledge.

Intrusion Detection Systems (IDS) monitor network or system activity to identify potentially suspicious behaviour. Traditional systems often rely on predefined signatures and rules to detect known threats. Although useful for recognising established attack patterns, these approaches may struggle to identify previously unseen or modified threats.

Machine learning-based IDS offer another approach by learning patterns from labelled network traffic and classifying observations into different categories. Their application to intrusion detection has been widely explored in cybersecurity research [1], [2]. However, a predicted category alone may not explain why an observation was flagged or which characteristics influenced the classification.

## 3.2 Explainability in Cybersecurity

Explainable Artificial Intelligence (XAI) aims to make the outputs and behaviour of AI models easier for people to understand. In cybersecurity, explanation methods can highlight the features that influence a model's prediction, providing additional context when examining a security alert [3], [8].

This study uses SHAP (SHapley Additive exPlanations), a feature-attribution method based on Shapley values from cooperative game theory. It estimates how individual input features contribute to a model's output and can be used to examine specific predictions as well as broader feature-importance patterns [13].

However, an explanation does not confirm that an attack has occurred. It describes how the model's output is influenced by the input features, but it does not establish the actual cause of a network event or guarantee that the classification is correct. The consistency and reliability of explanations therefore need to be examined rather than assumed.

## 3.3 Problem Statement

Cybersecurity detections may originate from multiple sources, while machine learning-based systems can produce predictions without clearly explaining the factors behind them. Without a way to bring these outputs together and present them clearly, security personnel may struggle to interpret alerts, recognise related activity and decide whether further investigation is required.

This difficulty is greater in organisations where security monitoring is handled by people with limited specialised cybersecurity knowledge. There is therefore a need to investigate how explainable machine learning can contribute to centralised cybersecurity monitoring and threat assessment, while making the basis and limitations of model predictions easier to understand.

## 3.4 Research Questions

The study is guided by the following primary research question:

**Can Explainable AI improve the interpretability of cyber threat detection and support human decision-making in a centralised cybersecurity monitoring and threat intelligence system?**

It also examines the following related question:

**How consistent are SHAP explanations across validation and test data, and across individual observations within the same traffic category?**

The second question reflects the explanation comparisons conducted in this phase. Explanation faithfulness is discussed as an important property of XAI, but it was not independently established through a formal faithfulness evaluation in the present experiments.

## 3.5 Scope of the Study

This work represents Phase One of ShadowScan, a broader research and development initiative focused on explainable, AI-assisted cybersecurity monitoring and decision support.

The first phase concentrates on dataset-based network intrusion detection using the RT-IoT2022 dataset. The work covers data preprocessing, training and comparison of machine learning classifiers, integration of SHAP with the selected model, and examination of its predictions and explanations.

A Streamlit-based prototype was developed to display model predictions and related explanation details. A screenshot-based survey was also conducted to explore how participants interpret selected predictions and explanations.

This phase does not include live network monitoring, automated incident response or a fully operational threat intelligence system. Instead, it serves as an initial proof of concept for the wider ShadowScan initiative.

## 3.6 Contributions

The work completed in Phase One includes:

1. Preparing the RT-IoT2022 dataset for multi-class network traffic classification.
2. Training and comparing Decision Tree and Random Forest classifiers.
3. Integrating SHAP to examine global, class-wise and individual predictions.
4. Analysing influential features, explanation consistency, feature relationships and recurring classification errors.
5. Developing a Streamlit prototype to present predictions and their explanations.
6. Conducting an initial screenshot-based survey to assess participants' understanding of selected model explanations.

# 4. Related Work and Literature Review

## 4.1 Machine Learning for Intrusion Detection

Machine learning has been widely studied as an approach to network intrusion detection. Unlike traditional signature-based systems, these models learn patterns from network traffic and use them to identify or classify suspicious activity. Ali et al. [1] provide a comprehensive review of machine learning strategies and considerations in intrusion detection systems. Other research has explored their application in specific environments, including digital education networks [2].

These studies provide a basis for understanding how machine learning can be applied to network security. However, predictive performance alone does not provide a complete picture of an intrusion detection system. A classifier may correctly identify a traffic category without showing which features influenced its decision. This limits the information available to security personnel when reviewing an alert or deciding whether further investigation is required.

## 4.2 Explainable AI in Cybersecurity

Explainable Artificial Intelligence has received attention as a way to make machine learning predictions easier to interpret. Zhang et al. [3] review the application of XAI methods in cybersecurity, while Arreche et al. [8] examine explainability frameworks in the context of network intrusion detection.

SHAP is one approach used to understand the contribution of individual features to model predictions. It can provide explanations at both individual and broader levels, helping examine why a model assigns a particular class to an observation.

However, generating an explanation does not automatically make it reliable. It is also necessary to examine whether the explanation represents the model's behaviour accurately and whether similar observations receive reasonably consistent explanations. Maseno et al. [9] address the reliability auditing of explanations in machine learning-based intrusion detection systems. Their work highlights the importance of evaluating explanations rather than assuming that they are dependable.

## 4.3 Human Understanding and Security Decision-Making

Cybersecurity monitoring involves more than identifying suspicious activity. Personnel must interpret available information, assess its importance and decide whether further action is required. This can be difficult when users have limited cybersecurity knowledge or when information is spread across different systems.

Research on human-centred XAI explores how explanations can be presented in ways that support users. Cecchi et al. [4] examine the relationship between explainable AI, cybersecurity, human-centred design and edge computing. Other work has explored human-centred intrusion detection [10] and explainable alert triage for Security Operations Centres [11].

Studies addressing cybersecurity awareness in education and security practices in healthcare [5], [6] provide a broader organisational context. They highlight the importance of considering the people, infrastructure and environments involved in managing security, rather than focusing only on technical detection. The study on data security perceptions in Indian hospitals [7] also relates to this wider context; however, only its publisher abstract was available for review.

## 4.4 Research Gap and Motivation

The reviewed literature covers machine learning-based intrusion detection, explanation methods, explanation reliability and human-centred security practices. These areas provide a foundation for studying how detection results can be made more understandable to people responsible for security assessment.

However, the usefulness of an explanation depends not only on what it displays but also on whether users can interpret it correctly and understand its limitations. A technically detailed explanation may still be difficult to use if its meaning is unclear or if users treat a model prediction as confirmation of an attack.

Phase One of ShadowScan explores these concerns through a dataset-based intrusion detection model integrated with SHAP. It examines model performance, feature contributions, explanation consistency and classification errors. A screenshot-based survey is also used to explore how participants interpret selected predictions and explanations.

The aim is to establish an initial foundation for a wider cybersecurity monitoring and decision-support system in which detection results and their explanations can be presented together to assist human assessment.

# 5. Research Methodology

## 5.1 Research Design

Phase One follows an experimental approach to investigate the use of Explainable Artificial Intelligence in network intrusion detection. A publicly available network traffic dataset was used to train and evaluate machine learning classifiers. SHAP was then integrated with the selected model to examine the features influencing its predictions.

The methodology also includes prototype development and a screenshot-based survey. These stages were used to explore how model predictions and explanations could be presented to users and how participants interpret the information displayed.

The overall process consists of the following stages:

1. Dataset selection and exploration.
2. Data preprocessing and preparation.
3. Model training and comparison.
4. Evaluation using validation and test data.
5. SHAP integration and explanation analysis.
6. Prototype development.
7. Human comprehension survey.

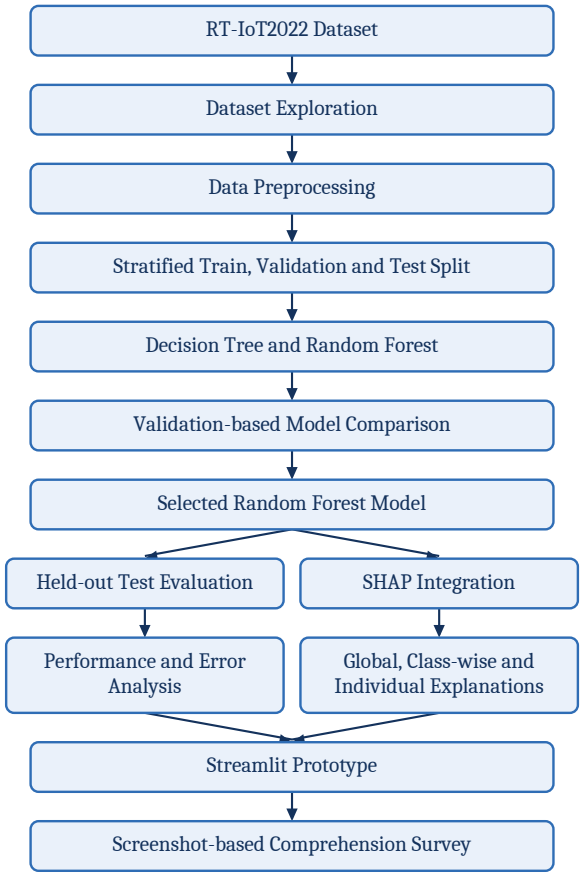



**Figure 5.1.** Research workflow

## 5.2 Dataset Selection and Description

The study uses the RT-IoT2022 dataset, obtained from the UCI Machine Learning Repository [12]. The dataset contains network traffic observations representing different types of normal and malicious activity. It was selected to support multi-class network traffic classification and the examination of model explanations across different traffic categories.

The dataset contains 123,117 observations and 85 original columns. The target variable, `Attack_type`, identifies the traffic category associated with each observation.

**Table 5.1.** Dataset characteristics

| Characteristic | Description |
|---|---|
| Dataset | RT-IoT2022 |
| Source | UCI Machine Learning Repository |
| Total observations | 123,117 |
| Original columns | 85 |
| Target variable | `Attack_type` |
| Target categories | 12 |
| Numerical predictors | 81 |
| Categorical predictors | 2 (`proto`, `service`) |
| Final features after preprocessing | 93 |

The dataset includes categories such as `DOS_SYN_Hping`, `MQTT_Publish`, `ARP_poisioning`, NMAP scans, `Thing_Speak` and `Wipro_bulb`. These categories represent different forms of network activity and provide a basis for evaluating the model's ability to distinguish between traffic types.

## 5.3 Data Preprocessing

Preprocessing was carried out to prepare the dataset for machine learning. The original data contained an index-like column, categorical variables and numerical values that required treatment before model training.

The following steps were performed:

1. **Removal of unnecessary columns:** The `Unnamed: 0` column was removed because it served as an index and was not required as a predictive feature.
2. **Categorical encoding:** The `proto` and `service` columns were converted into numerical representations using one-hot encoding.
3. **Handling infinite values:** Infinite values were replaced with missing values to allow further processing.
4. **Missing-value treatment:** Missing numerical values were filled using median imputation.
5. **Feature selection:** `VarianceThreshold` was applied to remove a constant feature that did not provide useful variation.
6. **Final feature preparation:** The resulting dataset contained 93 features for model training and evaluation.

These steps ensured that the input data was in a suitable format for the selected classifiers.

## 5.4 Dataset Splitting

The dataset was divided into training, validation and test partitions. A stratified split was used to maintain the distribution of the target categories across the partitions.

Initially, 80% of the observations were assigned to the training partition and the remaining 20% were reserved as the held-out test set. The initial training partition was then

divided further into model-training and validation data using an 80:20 split.

**Table 5.2. Dataset partitioning**

| Partition | Observations | Purpose |
|---|---|---|
| Model training | 78,794 | Training the classifiers |
| Validation | 19,699 | Comparing and selecting models |
| Held-out test | 24,624 | Evaluating the selected model |
| **Total** | **123,117** | |

The random state was set to 42 to maintain reproducibility.

The validation partition was used for model comparison, while the held-out test partition was kept separate for final performance evaluation.

**Preprocessing limitation:** The preprocessing objects were fitted on the initial 80% training partition before it was divided into model-training and validation data. As a result, the validation partition influenced some unsupervised preprocessing decisions, such as imputation and feature selection. The held-out test data was not used to fit these preprocessing steps. This limitation is considered when interpreting the validation results.

## 5.5 Model Development

Two supervised machine learning classifiers were selected for the experiment: Decision Tree and Random Forest. Both were trained to classify network traffic into the 12 categories present in the dataset.

**Decision Tree Classifier:** A Decision Tree learns a series of feature-based conditions to classify observations. It divides the data into branches based on selected features until it reaches a predicted category. Its relatively straightforward structure makes it useful as a baseline for comparison with an ensemble-based method.

**Random Forest Classifier:** Random Forest combines predictions from multiple decision trees. Each tree contributes to the final classification, allowing the model to capture different patterns in the data. For this experiment, the classifier was configured with 200 estimators, a random state of 42 and `n_jobs=-1`.

Both models were trained using the prepared model-training data and evaluated on the validation partition. Their performance was compared before selecting a model for further testing and explainability analysis.

## 5.6 Model Evaluation

The classifiers were evaluated using three performance metrics:

- **Accuracy:** Measures the proportion of correctly classified observations out of the total observations.
- **Macro F1-score:** Calculates the F1-score separately for each category and takes their unweighted average. This gives equal importance to each category, regardless of its size.
- **Balanced accuracy:** Calculates the average recall across all categories, helping account for differences in class distribution.

These metrics were considered together because the dataset contains an imbalance in the number of observations across its 12 categories. Accuracy alone may not adequately represent performance on categories with fewer observations.

The validation results were used to compare the two classifiers and select the model for the remaining stages of the study. The held-out test data was used for the final evaluation of the selected model.

## 5.7 Explainability Using SHAP

After model selection, SHAP (SHapley Additive exPlanations) was integrated with the Random Forest classifier using the `TreeExplainer` method.

SHAP assigns contribution values to input features to show how they influence a model's output. A positive contribution generally pushes the output towards the class being explained, while a negative contribution pushes it away from that class. The interpretation depends on the particular model output being examined.

The explanation analysis was conducted at three levels:

**Table 5.3. Levels of SHAP analysis**

| Analysis level | Purpose |
|---|---|
| Global | Identify features with the strongest overall influence on model predictions. |
| Class-wise | Examine which features contribute most to predictions for individual traffic categories. |
| Individual | Understand the feature contributions behind selected observations. |

In addition to feature importance, the study examined the consistency of explanations across observations and compared feature rankings between validation and test data. Misclassified observations were also reviewed to understand where the model's predictions differed from their actual categories.

SHAP explanations describe the behaviour of the trained model. They do not establish the actual cause of network activity or independently confirm whether an observation represents an attack. The feature-overlap analysis in this study provides a descriptive measure of explanation

consistency; it is not a formal test of explanation faithfulness.

### 5.8 Prototype Development

A Streamlit-based prototype was developed to present the model's predictions, performance information and SHAP-based explanations through a single interface.

The prototype contains four main sections:

**Table 5.4. Prototype modules**

| Section | Purpose |
|---|---|
| Traffic Inspector | Examine selected traffic observations and their predicted categories. |
| Model Performance | Display model evaluation results. |
| Global XAI Insights | Present overall and class-wise feature-importance information. |
| Misclassification Audit | Review observations where the predicted category differs from the actual category. |

The prototype uses the prepared research data and trained model artefacts. It demonstrates how detection results and explanations can be organised for examination through a common interface.

It is a local, dataset-based proof of concept. It does not perform live packet capture, continuous network monitoring or automated incident response.

### 5.9 Human Comprehension Survey

A screenshot-based survey was conducted using Google Forms to explore how participants interpret model predictions and SHAP explanations.

Participants were shown selected screenshots from the prototype and asked questions about the predicted traffic category, the features influencing a prediction, the meaning of SHAP contributions and the appropriate interpretation of a potential security alert.

The exercise was designed to examine whether participants could:

1. Identify the predicted traffic category.
2. Recognise the features supporting a prediction.
3. Interpret positive SHAP contributions.
4. Understand that a model explanation does not prove malicious activity.
5. Consider investigation before taking action.

The survey is ongoing. At the time of reporting, 11 responses had been received. The results are therefore preliminary and may change as additional responses are collected.

Since the exercise uses screenshots rather than an interactive system, it evaluates participants' understanding of selected examples. It does not establish whether XAI improves operational decision-making or the effectiveness of a live cybersecurity monitoring system.

## 6. Results, Analysis and Prototype Implementation

This section presents the experimental results obtained during Phase One of ShadowScan, followed by an analysis of the model's explanations and the implementation of the Streamlit prototype. It covers model performance, global and class-wise SHAP analysis, explanation consistency, classification errors, feature correlations and application verification.

### 6.1 Model Comparison

Decision Tree and Random Forest classifiers were evaluated using the validation dataset. Accuracy, macro F1-score and balanced accuracy were considered to compare their performance across the 12 traffic categories.

**Table 6.1. Validation performance comparison**

| Evaluation metric | Decision Tree | Random Forest |
|---|---|---|
| Accuracy | 99.71% | 99.81% |
| Macro F1-score | 95.87% | 97.18% |
| Balanced accuracy | 95.67% | 95.29% |

Random Forest achieved higher accuracy and macro F1-score, while Decision Tree recorded slightly higher balanced accuracy. Considering the overall results, Random Forest was selected for further evaluation and SHAP integration.

The selection was not based on accuracy alone. The macro F1-score was also considered because it gives equal importance to each traffic category, irrespective of the number of observations.

### 6.2 Random Forest Test Results

The selected Random Forest classifier was evaluated on the held-out test partition containing 24,624 observations. This partition was kept separate from model training and validation-based model selection.

**Table 6.2. Random Forest performance on the held-out test dataset**

| Evaluation metric | Result |
|---|---|
| Total test observations | 24,624 |
| Correct predictions | 24,581 |
| Incorrect predictions | 43 |
| Accuracy | 99.83% |
| Macro F1-score | 96.22% |
| Balanced accuracy | 96.57% |

The model correctly classified 24,581 observations, achieving an accuracy of 99.83%. The macro F1-score and balanced accuracy provide a further view of its performance across the different categories.

However, the dataset has a considerable class imbalance. DOS_SYN_Hping accounts for approximately 76.9% of the test observations, while some categories have very few samples. In particular, Metasploit_Brute_Force_SSH has 7 test observations and NMAP_FIN_SCAN has only 6.

Therefore, overall accuracy should not be considered in isolation. The results for less frequent categories also need to be examined when assessing the model's performance.

## 6.3 Global SHAP Feature Analysis

SHAP TreeExplainer was used to examine the contribution of input features to the Random Forest model's predictions. Global feature importance was calculated using the mean absolute SHAP value across all test observations and output classes.

**Table 6.3. Top 10 global SHAP features on the held-out test dataset**

| Rank | Feature | Mean absolute SHAP value |
|---|---|---|
| 1 | id.resp_p | 0.007281 |
| 2 | fwd_pkts_payload.avg | 0.006901 |
| 3 | fwd_pkts_payload.min | 0.006240 |
| 4 | fwd_pkts_payload.max | 0.004749 |
| 5 | fwd_subflow_bytes | 0.004433 |
| 6 | fwd_pkts_payload.tot | 0.003620 |
| 7 | service_- | 0.003442 |
| 8 | flow_iat.min | 0.002524 |
| 9 | active.avg | 0.002509 |
| 10 | flow_duration | 0.002489 |

Destination port information (id.resp_p) and forward packet payload measurements appeared among the highest-ranked features. Other influential features included subflow bytes, service information and flow timing measurements.

The global feature ranking was also compared between the validation and test partitions. The comparison covered all 93 features.

**Table 6.4. Validation and test feature-ranking comparison**

| Comparison measure | Result |
|---|---|
| Features compared | 93 |
| Spearman rank correlation | 0.9999 |
| Top-10 feature overlap | 100% |

The rankings were highly similar across the two partitions, with all ten highest-ranked features appearing in both. This indicates that the global feature-importance ranking remained similar across the validation and test data.

Mean absolute SHAP values represent the magnitude of feature contributions. They do not indicate whether a feature consistently increases or decreases a particular class prediction.

## 6.4 Class-wise SHAP Analysis

Global feature importance provides an overall view of the model, but different traffic categories may rely on different combinations of features. Class-wise SHAP analysis was therefore performed on the held-out test dataset.

For each category, observations were grouped using their actual labels. The mean absolute SHAP values were then calculated for the corresponding class output. This analysis includes observations that were misclassified, rather than considering only correctly predicted records.

Some notable findings from the class-wise results were:

- **MQTT_Publish:** service_mqtt ranked second among the most influential features.
- **Metasploit_Brute_Force_SSH:** service_ssh ranked second.
- **NMAP_XMAS_TREE_SCAN:** fwd_URG_flag_count and flow_FIN_flag_count ranked first and second, respectively.
- **NMAP_FIN_SCAN:** flow_FIN_flag_count appeared among its top five features.
- **DOS_SYN_Hping:** Destination port and forward packet payload measurements were among the leading features.

These results show that the model used different feature combinations when classifying different categories. For example, service-related features were relevant to MQTT and SSH traffic, while TCP flag-related features appeared prominently in the NMAP_XMAS_TREE_SCAN category.

The class-wise findings provide more specific information about individual traffic categories than the global ranking alone. However, results for categories with very few test observations should be interpreted cautiously.

## 6.5 Explanation Consistency Analysis

The study also examined how closely individual explanations followed the overall feature patterns of their respective traffic categories.

For this analysis, the ten highest-ranked features from each individual test observation were compared with the ten highest-ranked features from the aggregate explanation for its actual category.

The overlap was calculated as:

**Top-10 Feature Overlap (%) = (Number of common features / 10) × 100**

The overlap was averaged across all observations belonging to each category.

**Table 6.5. Intra-class explanation consistency on the held-out test dataset**

| Traffic category | Average top-10 overlap | Test support |
|---|---|---|
| ARP_poisioning | 69.87% | 1,550 |
| DDOS_Slowloris | 74.49% | 107 |
| DOS_SYN_Hping | 99.71% | 18,932 |
| MQTT_Publish | 98.77% | 829 |
| Metasploit_Brute_Force_SSH | 72.86% | 7 |
| NMAP_FIN_SCAN | 88.33% | 6 |
| NMAP_OS_DETECTION | 96.03% | 400 |
| NMAP_TCP_scan | 91.35% | 200 |
| NMAP_UDP_SCAN | 81.06% | 518 |
| NMAP_XMAS_TREE_SCAN | 93.66% | 402 |
| Thing_Speak | 82.91% | 1,622 |
| Wipro_bulb | 64.31% | 51 |

The results varied across categories. `DOS_SYN_Hping` and `MQTT_Publish` showed high overlap, while `Wipro_bulb` and `ARP_poisioning` showed lower values.

For example, an overlap of 69.87% indicates that, on average, approximately seven of the ten highest-ranked features in an individual explanation were also present in the aggregate top-ten feature set for that category.

This measure describes similarity in feature rankings within a category. It does not establish that every explanation is reliable or that the model would produce the same explanations under different datasets or conditions.

The results for `Metasploit_Brute_Force_SSH` and `NMAP_FIN_SCAN` require particular caution because their test supports were only 7 and 6 observations, respectively.

## 6.6 Misclassification Analysis

The Random Forest model produced 43 incorrect predictions on the held-out test dataset. These were examined to identify recurring patterns between actual and predicted categories.

The most frequent confusion occurred between `Thing_Speak` and `ARP_poisioning`.

**Table 6.6. Most frequent test-set misclassifications**

| Actual category | Predicted category | Observations |
|---|---|---|
| Thing_Speak | ARP_poisioning | 13 |
| ARP_poisioning | Thing_Speak | 10 |

These two directions account for 23 of the 43 incorrect predictions. The remaining errors were distributed across other category pairs.

SHAP contributions were also examined for recurring errors to identify the features that supported the predicted category and those that contributed towards the actual category.

This analysis helps examine why certain observations were assigned an incorrect label. It also demonstrates the value of reviewing explanations alongside model predictions, rather than relying only on the final classification.

## 6.7 Feature Correlation Analysis

A Spearman correlation analysis was performed on the 20 highest-ranked global SHAP features using the held-out test data. Feature pairs with an absolute correlation coefficient of 0.85 or higher were considered strongly correlated.

Several flow timing measurements showed very high correlations. Similar relationships were also observed between forward packet payload measurements.

This is relevant when interpreting SHAP results because correlated features may contain overlapping information. Their contributions can be distributed across related features, making it difficult to interpret the importance of each one independently.

The correlation analysis identifies statistical relationships between features. It does not establish a causal relationship between them.

*Note for final technical verification: retain a numerical correlation table only after checking each coefficient against the original notebook output or exported correlation results. The current project archive does not contain a separate correlation-results CSV, so this report describes the observed relationships without presenting unverified coefficients.*

## 6.8 Prototype Development and Implementation

Following the model and explainability experiments, a Streamlit-based prototype was developed to bring the different outputs together in an accessible interface.

The prototype demonstrates how network traffic records, model predictions and SHAP explanations can be presented through a common application. It is designed as a local, dataset-based decision-support tool, following the principle of **"Decision support, not decision replacement."**

### 6.8.1 System Workflow

The implemented prototype follows the workflow shown in Figure 6.1.

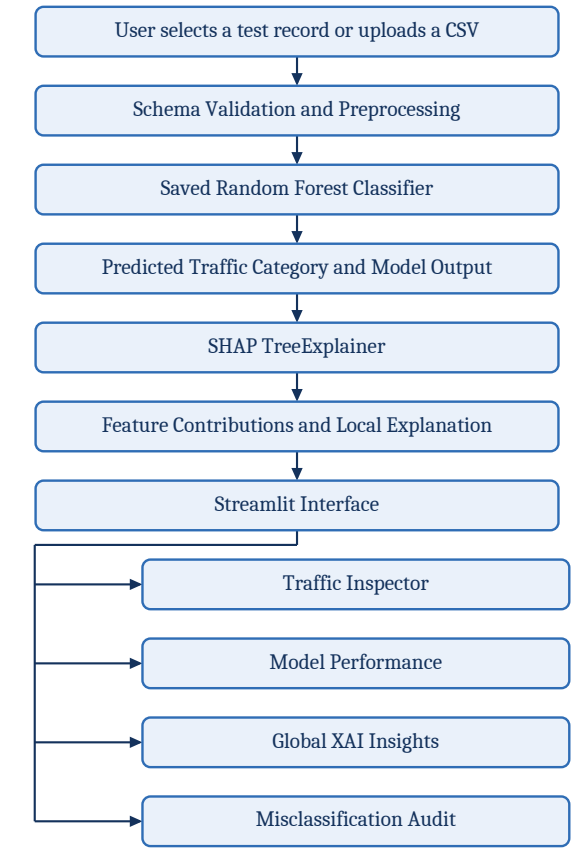



**Figure 6.1.** Prototype workflow

### 6.8.2 Application Modules

The prototype contains four main modules.

**1. Traffic Inspector**

This module allows users to inspect selected observations from the held-out test dataset or upload a CSV file containing network flow records that follow the expected dataset schema.

Its functions include:

- Displaying the predicted traffic category and class probabilities.
- Presenting model confidence information.
- Showing individual SHAP feature contributions.
- Displaying waterfall-based explanation visualisations.
- Providing human-readable summaries of features supporting or opposing a prediction.
- Presenting batch prediction results and CSV export for uploaded records.

**2. Model Performance**

This module presents the Random Forest evaluation results on the held-out test dataset. It includes:

- An interactive 12-class confusion matrix.
- Raw and normalised confusion matrix views.
- Class-wise precision, recall and F1-score.
- Low-support category warnings.
- Overall benchmark metrics.

**3. Global XAI Insights**

This module presents broader explanation results, including:

- Global SHAP feature rankings.
- Class-wise feature importance.
- Positive and negative feature contributions.
- Intra-class explanation consistency.
- Validation and test feature-ranking comparison.
- Correlation analysis of globally important features.

**4. Misclassification Audit**

This module focuses on observations where the model's predicted category differs from the actual category. It presents recurring misclassification patterns and examines SHAP contributions associated with the most frequent errors.

This allows users to inspect the features influencing incorrect predictions and compare their contributions towards the predicted and actual categories.

### 6.8.3 Implementation and Verification

The application was organised into separate modules for preprocessing, model management, SHAP explanations, evaluation and user interface components.

The trained Random Forest model and preprocessing objects were saved for reuse. This allows the prototype to process compatible input records without retraining the classifier each time.

An end-to-end verification script, `verify_app.py`, was also included to check the implementation.

The verification covered six main areas:

**Table 6.7. Application verification**

| Verification area | Purpose |
|---|---|
| Research file preservation | Check that the original dataset, notebook and preprocessing files remain intact. |
| Model verification | Confirm that the saved model reproduces the recorded test metrics. |
| Preprocessing verification | Check that inference produces the expected 93 features. |
| SHAP verification | Test local explanations and narrative generation. |
| Evaluation data verification | Confirm that the saved research CSV files load and contain the expected records. |
| Interface verification | Compile the main Streamlit application and its four modules. |

All six verification suites passed during the recorded implementation checks. This confirmed that the saved model, preprocessing pipeline, explanation components, evaluation files and application modules operated as expected in the verification environment.

The model artefacts and verification environment used different scikit-learn versions (1.9.1 and 1.8.0, respectively). Although the verification checks passed, consistent dependency versions are recommended for reproducibility.

### 6.9 Limitations of the Experimental Results and Prototype

The findings should be considered within the scope of the experiments conducted.

1. **Dataset limitation:** The model was evaluated using RT-IoT2022. Its performance on this dataset does not guarantee similar results on live network traffic or in other environments.
2. **Class imbalance:** Some categories have very few observations, particularly `Metasploit_Brute_Force_SSH` and `NMAP_FIN_SCAN`. Their performance and explanation patterns require cautious interpretation.
3. **Preprocessing limitation:** Some preprocessing objects were fitted on the initial 80% training partition before it was divided into model-training and validation subsets. Consequently, validation data influenced certain unsupervised preprocessing decisions. The held-out test data was not used to fit these objects.
4. **Explanation limitation:** SHAP describes the model's feature contributions. It does not establish causation, confirm malicious activity or guarantee that a prediction is correct.
5. **Feature correlation:** Strong relationships between input features may affect how SHAP contributions are distributed and interpreted.
6. **Prototype scope:** The application operates on prepared benchmark records and compatible CSV uploads. It does not include live packet capture, continuous network monitoring or automated incident response.
7. **Software environment:** The saved model artefacts and verification environment used different scikit-learn versions. Although the verification checks passed, using consistent dependency versions is recommended for reproducibility.

The screenshot-based human comprehension survey remains ongoing. Its findings need to be considered separately before drawing conclusions about participants' understanding or the usefulness of explanations for human decision-making.

## 7. Human Comprehension Survey and Findings

### 7.1 Survey Overview

A screenshot-based survey was conducted to explore how participants interpret machine learning predictions and SHAP explanations presented through the ShadowScan prototype.

Participants were shown selected screenshots and asked questions about the predicted traffic category, influential features, SHAP contributions and the appropriate response to a potential security alert.

The survey aimed to examine whether participants could understand the information presented, identify the features supporting a prediction and recognise the limitations of model explanations.

At the time of reporting, 11 responses had been received. The survey remains open, and the findings presented here are preliminary.

### 7.2 Survey Results

The responses were analysed using the number and percentage of participants who selected the expected answer.

**Table 7.1. Preliminary survey findings**

| Assessment area | Correct responses | Percentage |
|---|---|---|
| Identifying the predicted category – Scenario 1 | 10/11 | 90.9% |
| Identifying the predicted category – Scenario 2 | 11/11 | 100% |
| Identifying the predicted category – Scenario 3 | 10/11 | 90.9% |
| Identifying a supporting feature – Scenario 2 | 11/11 | 100% |
| Understanding a positive SHAP contribution | 9/11 | 81.8% |
| Understanding that SHAP does not prove malicious activity | 8/11 | 72.7% |
| Selecting investigation before taking action | 9/11 | 81.8% |

Participants generally identified the predicted categories correctly across the three scenarios. All 11 participants also identified a supporting feature in the second scenario.

Understanding of SHAP contributions was comparatively lower. Nine participants correctly interpreted a positive SHAP contribution, while eight recognised that an explanation does not independently establish whether an observation is malicious.

For the decision-making question, nine participants selected investigation before taking action. This indicates that most respondents recognised the need for further examination rather than treating a model prediction as sufficient grounds for an immediate response.

### 7.3 Interpretation of Survey Findings

The preliminary responses suggest that participants were able to interpret several elements of the displayed predictions and explanations. In particular, identifying predicted categories and supporting features appeared relatively straightforward in the selected examples.

However, some participants had difficulty interpreting the limitations of SHAP explanations. This is important because a feature contribution describes how the model arrived at a prediction; it does not confirm that the predicted activity is an actual attack.

The responses also suggest that presenting an explanation alongside a prediction may provide users with additional information when considering a security alert. However, the survey does not establish that XAI improves decision-making or that participants would respond similarly in a real cybersecurity environment.

### 7.4 Survey Limitations

The survey findings must be interpreted with caution for the following reasons:

- The current sample consists of only 11 responses.
- Participants examined screenshots rather than interacting directly with the application.
- The questions covered selected examples and do not represent every possible network traffic category or security scenario.
- The exercise assessed responses to specific questions rather than measuring changes in understanding before and after using XAI.
- The results do not establish the effectiveness of the prototype in an operational security environment.

The survey will remain open for additional responses. The findings and percentages may be updated in the final version of the report.

## 8. Discussion

### 8.1 Model Performance and Intrusion Detection

The experimental results show that the Random Forest classifier achieved high classification performance on the RT-IoT2022 dataset. Its test accuracy was 99.83%, with a macro F1-score of 96.22% and balanced accuracy of 96.57%.

The comparison with Decision Tree showed that Random Forest performed better in accuracy and macro F1-score, although it recorded slightly lower balanced accuracy. This demonstrates why multiple evaluation metrics are necessary when working with an imbalanced dataset.

The class distribution also affects the interpretation of the results. `DOS_SYN_Hping` represents approximately 76.9% of the test observations, while some categories contain very few records. Although the overall results are strong, performance on minority categories must be considered separately.

The model therefore provides a useful foundation for the explainability experiments, but its performance should not be generalised to other network environments without further testing.

### 8.2 Role of SHAP in Understanding Predictions

The integration of SHAP provided additional information about the features influencing the Random Forest predictions.

Global analysis identified destination port information and forward packet payload measurements among the most influential features. Class-wise analysis showed that different

traffic categories were associated with different feature patterns. Individual explanations provided further detail about the contributions behind selected predictions.

This demonstrates how feature-attribution methods can make model outputs easier to examine. Instead of viewing only the predicted category, a user can inspect which input characteristics contributed towards the classification.

However, SHAP explanations must be interpreted within the limits of the model and the dataset. They describe feature contributions to a model output and do not establish the actual cause of network activity or confirm whether an attack has occurred.

### 8.3 Consistency and Reliability of Explanations

The study examined explanation consistency through two comparisons: the similarity of global feature rankings between validation and test data, and the overlap between individual and class-level feature rankings.

The global comparison showed a Spearman rank correlation of 0.9999, with all ten highest-ranked features appearing in both partitions. The intra-class overlap analysis, however, showed differences between traffic categories.

These findings suggest that the overall feature-importance ranking remained similar across the two partitions, while individual observations did not always rely on exactly the same set of influential features.

The overlap measure provides a descriptive assessment of feature-ranking similarity. It is not a complete reliability audit and does not independently establish explanation faithfulness, robustness or stability under changes to the input data.

Further experiments would be required to evaluate these properties more thoroughly.

### 8.4 Misclassification and the Importance of Human Review

The analysis identified recurring confusion between `Thing_Speak` and `ARP_poisioning`, accounting for 23 of the 43 test errors across the two prediction directions.

Examining these errors alongside their SHAP contributions provides additional information about why the model favoured an incorrect category. This is useful when reviewing the limitations of a classifier, as the predicted label alone does not reveal the features associated with the mistake.

The findings support the need for human review when interpreting security detections. A model prediction can help identify activity that requires attention, but it should not replace further investigation or the judgement of responsible personnel.

### 8.5 Human Understanding and Decision Support

The preliminary survey results indicate that participants could identify the predicted categories and supporting features in the selected examples. However, fewer participants correctly interpreted the limitations of SHAP explanations.

This highlights an important distinction between displaying an explanation and ensuring that users understand it correctly. An explanation may contain useful technical information, but its presentation must also make its meaning and limitations clear.

The survey provides an initial view of how participants interpret selected predictions. Since the sample is small and the exercise is screenshot-based, it cannot establish whether the prototype improves real-world decision-making.

### 8.6 Contribution Towards the Wider ShadowScan Initiative

Phase One establishes an initial technical foundation for ShadowScan by combining network intrusion detection, model explanations, error analysis and a prototype interface.

The work demonstrates how predictions and their explanations can be organised in one application for examination. It also identifies areas that require further investigation, including explanation reliability, event correlation, minority-category performance and human interpretation.

The prototype is not a complete centralised cybersecurity monitoring system. Instead, it represents an early proof of concept towards the wider objective of bringing security information together and assisting personnel in understanding and prioritising potential threats.

## 9. Limitations and Future Scope

### 9.1 Limitations

Although Phase One produced encouraging experimental results, several limitations remain.

**1. Dataset dependency**

The experiments were conducted using the RT-IoT2022 dataset. The model has not been evaluated on live network traffic or multiple independent datasets. Its ability to generalise to different network environments therefore remains untested.

**2. Class imbalance**

The dataset contains a large difference in the number of observations across categories. Some minority categories have very limited test support, which restricts the conclusions that can be drawn about their classification performance.

**3. Preprocessing procedure**

Preprocessing objects were fitted on the initial training partition before it was divided into model-training and validation subsets. Consequently, validation data influenced some unsupervised preprocessing decisions. The held-out test data was not used to fit these preprocessing objects.

**4. Explainability evaluation**

The study examined global feature rankings and intra-class feature overlap. These measures provide useful information about explanation consistency but do not constitute a complete evaluation of faithfulness, robustness or reliability.

**5. Feature correlation**

Several input features showed strong correlations. This can affect the distribution of SHAP contributions between related features and makes individual feature importance more difficult to interpret independently.

**6. Human evaluation**

The screenshot-based survey currently has 11 responses. It does not involve a controlled comparison, direct interaction with the prototype or testing in a real security monitoring environment.

**7. Prototype functionality**

The current application works with prepared research data and compatible CSV inputs. It does not include live packet capture, continuous monitoring, integration with external security systems or automated incident response.

### 9.2 Future Scope

Phase One provides several directions for further research and development.

**1. Improved model evaluation**

Future experiments can evaluate the model using additional datasets and different network environments. Further attention can also be given to minority categories and methods for handling class imbalance.

**2. Detailed explanation reliability testing**

SHAP explanations can be examined using additional evaluation methods to study faithfulness, stability and sensitivity to changes in input features. This would provide a more complete understanding of explanation reliability.

**3. Event correlation**

A future phase can explore how separate detections and network events can be connected to identify related activity. This would move beyond examining individual observations towards understanding broader patterns.

**4. Risk assessment and prioritisation**

The system can be extended to organise detected events according to their potential importance. The objective would be to assist responsible personnel in deciding which events require closer examination.

**5. Human-centred interface development**

The prototype can be evaluated with a larger and more varied participant group. Further work can examine whether different explanation formats improve comprehension and whether users can distinguish between model predictions and confirmed security incidents.

**6. Edge-assisted and local-first architecture**

Future development can explore edge-based processing and local-first deployment to support organisations with limited infrastructure and cybersecurity resources.

**7. Response assistance**

The broader ShadowScan initiative can explore ways to provide response recommendations and investigation guidance. Any response-related functionality should retain human oversight, following the principle of decision support rather than decision replacement.

## 10. Conclusion

Phase One of ShadowScan investigated the use of Explainable Artificial Intelligence in network intrusion detection and explored its potential to support human understanding of cybersecurity predictions.

The study used the RT-IoT2022 dataset to train and compare Decision Tree and Random Forest classifiers. Random Forest was selected for further analysis and achieved 99.83% accuracy on the held-out test dataset, with a macro F1-score of 96.22% and balanced accuracy of 96.57%.

SHAP was integrated with the selected model to examine global, class-wise and individual feature contributions. The analysis provided additional information about the characteristics influencing predictions, recurring classification errors and similarities between explanations. It also highlighted the importance of considering class imbalance, correlated features and the limitations of feature-attribution methods.

A Streamlit-based prototype was developed to bring model predictions, performance results and explanation-related information together through four modules. An initial screenshot-based survey was also conducted to examine how participants interpret selected predictions and explanations. The preliminary responses indicate that participants could identify several displayed results, although some had difficulty understanding the limitations of SHAP explanations.

The findings demonstrate that explainability methods can provide additional insight into the behaviour of machine

learning-based intrusion detection models. However, the experiments do not establish that XAI independently improves operational decision-making or that the prototype is suitable for live cybersecurity monitoring.

Overall, Phase One establishes an initial proof of concept for the wider ShadowScan initiative. It provides a foundation for further work on explanation reliability, event correlation, risk prioritisation and human-centred cybersecurity monitoring, while retaining human judgement as an essential part of security assessment.

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